

Day 0: Workstation Set-up

Real-World DevOps Projects

Field	Details
Author	Rohan Aviral
Series	Real-World DevOps Projects (20 projects)
This part	Day 0: preparing the machine before Project 1
Machine	MacBook Pro (Intel), macOS 26.7.1, 16 GB RAM
GitHub	https://github.com/Aviral-Rohan

1. Goal

Before starting a 20-project series, the workstation has to be ready: every tool installed, Docker sized correctly, and a multi-node Kubernetes cluster proven to work locally. This document records what I installed, what went wrong on an Intel Mac, and how I got around it.

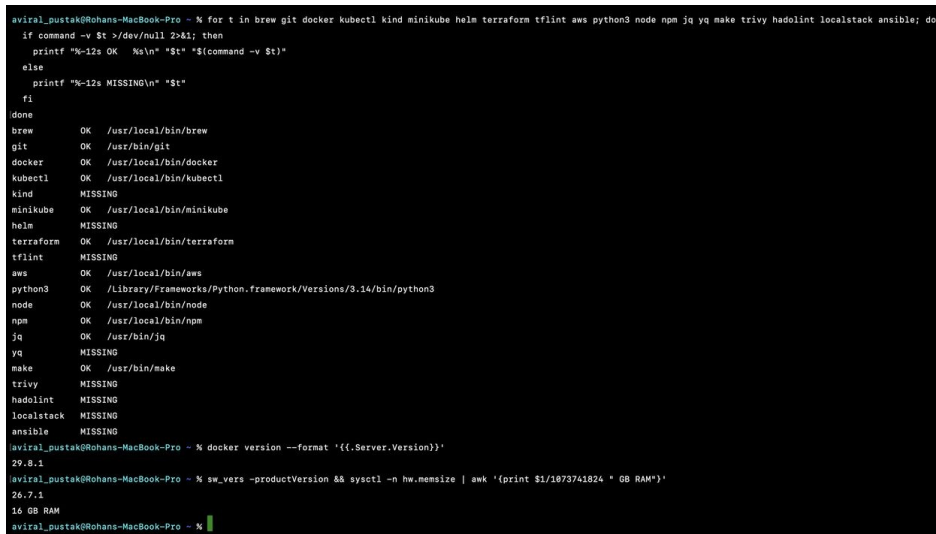
The tool set

Tool	Purpose	Installed with
git, docker, kubectl, minikube, terraform, aws, python3, node, npm, jq, make, brew	Already present	-
kind	Multi-node Kubernetes inside Docker	Homebrew
helm	Packaging Kubernetes applications	Homebrew
yq	Reading and editing YAML on the command line	Homebrew
trivy	Scanning images for vulnerabilities	Homebrew
hadolint	Linting Dockerfiles	Homebrew
tflint	Linting Terraform	Release binary from GitHub
localstack, awslocal	A local AWS simulator for free practice	pip
ansible	Configuration management	pip

2. Step 1: Check What Is Already Installed

The first thing to do on any new machine or server is find out what is already there. A short loop over command -v prints OK or MISSING for each tool. The same loop is in check-tools.sh in this folder.

```
for t in brew git docker kubectl kind minikube helm terraform tflint aws \
    python3 node npm jq yq make trivy hadolint localstack ansible; do
    if command -v $t >/dev/null 2>&1; then
        printf "%-12s OK    %s\n" "$t" "$(command -v $t)"
    else
        printf "%-12s MISSING\n" "$t"
    fi
done
```



```
aviral_pustak@Rohans-MacBook-Pro ~ % for t in brew git docker kubectl kind minikube helm terraform tflint aws python3 node npm jq yq make trivy hadolint localstack ansible; do
if command -v $t >/dev/null 2>&1; then
    printf "%-12s OK    %s\n" "$t" "$(command -v $t)"
else
    printf "%-12s MISSING\n" "$t"
fi
done
brew      OK    /usr/local/bin/brew
git       OK    /usr/bin/git
docker   OK    /usr/local/bin/docker
kubectl  OK    /usr/local/bin/kubectl
kind      MISSING
minikube  OK    /usr/local/bin/minikube
helm      MISSING
terraform OK    /usr/local/bin/terraform
tflint    MISSING
aws       OK    /usr/local/bin/aws
python3   OK    /Library/Frameworks/Python.framework/Versions/3.14/bin/python3
node      OK    /usr/local/bin/node
npm       OK    /usr/local/bin/npm
jq        OK    /usr/bin/jq
yq        MISSING
make      OK    /usr/bin/make
trivy     MISSING
hadolint  MISSING
localstack MISSING
ansible  MISSING
aviral_pustak@Rohans-MacBook-Pro ~ % docker version --format '{{.Server.Version}}'
29.8.1
aviral_pustak@Rohans-MacBook-Pro ~ % sw_vers -productVersion && sysctl -n hw.memsize | awk '{print $1/1073741824 " GB RAM"}'
26.7.1
16 GB RAM
aviral_pustak@Rohans-MacBook-Pro ~ %
```

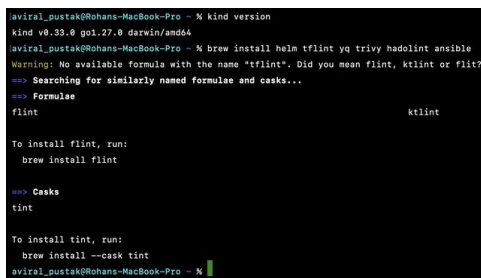
Screenshot 1: Starting point: 12 of 20 tools present, Docker 29.8.1 running, 16 GB RAM

3. Step 2: Install the Missing Tools

Eight tools were missing. On an Apple Silicon Mac this is one brew command. On an Intel Mac it turned into four separate problems, which are the useful part of this write-up.

Problem 1: a package Homebrew could not find

kind installed in seconds. The next command listed six tools, and Homebrew did not know one of them (tflint). Because one name was wrong, none of the six were installed.



```
aviral_pustak@Rohans-MacBook-Pro ~ % kind version
kind v0.33.0 go1.27.0 darwin/amd64
aviral_pustak@Rohans-MacBook-Pro ~ % brew install helm tflint yq trivy hadolint ansible
Warning: No available formula with the name "tflint". Did you mean flint, ktlint or flit?
=> Searching for similarly named formulae and casks...
=> Formulae
flint                                ktlint                               flit
To install flint, run:
  brew install flint
=> Casks
tint
To install tint, run:
  brew install --cask tint
aviral_pustak@Rohans-MacBook-Pro ~ %
```

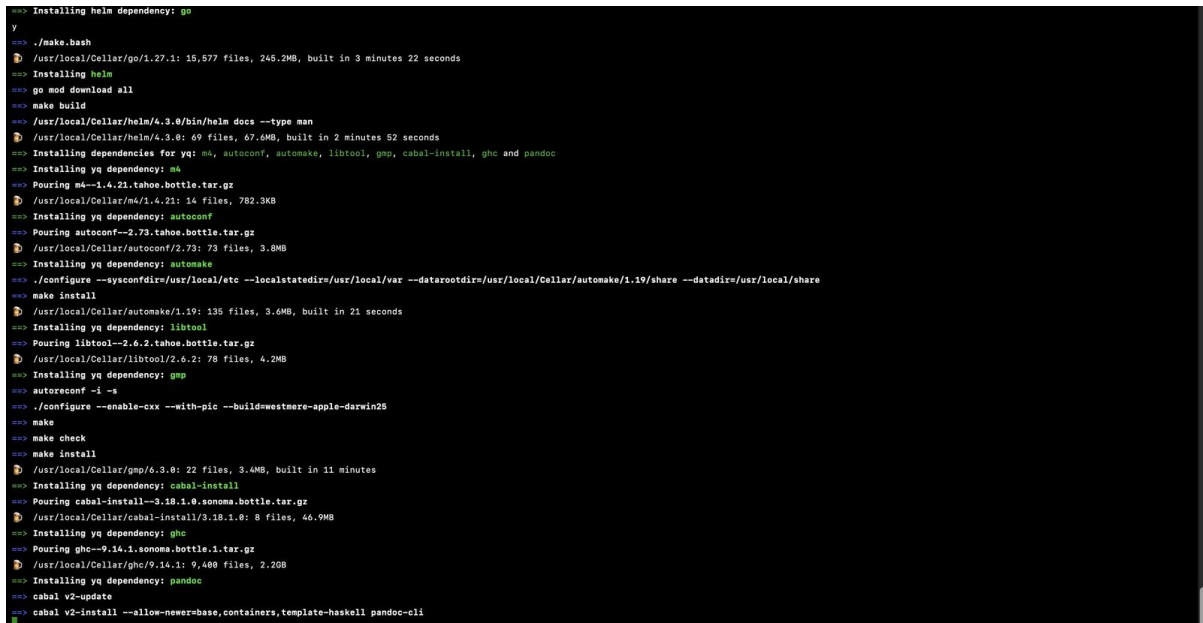
Screenshot 2: kind installed; then 'No available formula with the name tflint' stops the whole command

Fix: I removed tflint from the brew command and installed it as a ready-made binary from its GitHub releases page.

```
cd /tmp && curl -LO
https://github.com/terraform-linters/tflint/releases/latest/download/
tflint_darwin_amd64.zip
unzip -o tflint_darwin_amd64.zip && mv tflint /usr/local/bin/ && tflint --
version
```

Problem 2: Homebrew compiling everything from source

Homebrew no longer provides ready-made packages (bottles) for Intel Macs, so it builds from source when no bottle exists. Installing yq pulled in the Haskell compiler and started compiling pandoc. One small library took 11 minutes.

A terminal window showing the output of Homebrew commands. The process starts with installing the 'helm' dependency for 'yq'. It then proceeds to install 'yq' itself, which triggers the installation of several dependencies: 'mk', 'autoconf', 'automake', 'libtool', 'gmp', 'cabal-install', 'ghc', and 'pandoc'. The logs show the progress of each step, including downloading bottles for some packages and building from source for others. For example, 'gmp' is built from source and takes 11 minutes. The final step is installing 'cabal v2' and 'pandoc'.

Screenshot 3: Lines with `./configure` and `make` are source builds; 'Pouring' lines are fast ready-made downloads

The plan for ansible was worse: it was going to compile CMake, Python, LLVM and Rust, which is many hours on a laptop. I let Homebrew finish helm, yq, trivy and hadolint, then pressed Ctrl + C when it reached the ansible dependencies and installed ansible with pip instead. That took about a minute.

```
➡ make
^C

This is a Tier 3 configuration:
https://docs.brew.sh/Support-Tiers#tier-3
You can report issues with Tier 3 configurations to Homebrew/* repositories!

Read the above document before opening any issues or PRs.
aviral_pustak@Rohans-MacBook-Pro ~ % python3 -m pip install --user ansible
Collecting ansible
  Downloading ansible-14.5.0-py3-none-any.whl.metadata (8.1 kB)
Collecting ansible-core==2.21.5 (from ansible)
  Downloading ansible-core-2.21.5-py3-none-any.whl.metadata (7.8 kB)
Requirement already satisfied: Jinja2>=3.1.0 in /Library/Frameworks/Python.framework/Versions/3.14/lib/python3.14/site-packages (from ansible-core==2.21.5->ansible) (3.1.6)
Requirement already satisfied: PyYAML>=5.1 in /Library/Python/3.14/lib/python/site-packages (from ansible-core==2.21.5->ansible) (6.0.3)
Requirement already satisfied: cryptography in /Library/Python/3.14/lib/python/site-packages (from ansible-core==2.21.5->ansible) (50.0.2)
Requirement already satisfied: packaging in /Library/Python/3.14/lib/python/site-packages (from ansible-core==2.21.5->ansible) (26.3)
Collecting resolvelib<3.0.0,>=0.8.0 (from ansible-core==2.21.5->ansible)
  Downloading resolvelib-1.2.1-py3-none-any.whl.metadata (3.7 kB)
Requirement already satisfied: MarkupSafe>=2.0 in /Library/Frameworks/Python.framework/Versions/3.14/lib/python3.14/site-packages (from Jinja2>=3.1.0->ansible-core==2.21.5->ansible) (3.0.3)
Requirement already satisfied: cffi>=2.0.0 in /Library/Python/3.14/lib/python/site-packages (from cryptography->ansible-core==2.21.5->ansible) (2.1.1)
Requirement already satisfied: pycparser in /Library/Python/3.14/lib/python/site-packages (from cffi>=2.0.0->cryptography->ansible-core==2.21.5->ansible) (3.0)
Downloading ansible-14.5.0-py3-none-any.whl (51.4 MB)
  81.4/81.4 MB 1.4 MB/s 0:00:31
Downloading ansible-core-2.21.5-py3-none-any.whl (2.5 MB)
  2.5/2.5 MB 4.4 MB/s 0:00:00
Downloading resolvelib-1.2.1-py3-none-any.whl (18 kB)
Installing collected packages: resolvelib, ansible-core, ansible
  ____
 1/3 [ansible-core] WARNING: The scripts ansible, ansible-config, ansible-console, ansible-doc, ansible-galaxy, ansible-inventory, ansible-playbook, ansible-pull, ansible-test and
ansible-vault are installed in '/Users/aviral_pustak/Library/Python/3.14/bin' which is not on PATH.
  ____
Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.
  ____
2/3 [ansible] WARNING: The script ansible-community is installed in '/Users/aviral_pustak/Library/Python/3.14/bin' which is not on PATH.
  ____
Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.
Successfully installed ansible-14.5.0 ansible-core-2.21.5 resolvelib-1.2.1

[notice] A new release of pip is available: 26.1.2 -> 26.2.1
[notice] To update, run: pip install --upgrade pip
aviral_pustak@Rohans-MacBook-Pro ~ %
```

Screenshot 4: Build stopped with Ctrl + C; pip installs ansible 14.5.0 in about a minute

Problem 3: 'command not found' right after a successful install

pip installed LocalStack without errors, but the shell could not find the localstack command. The warning in the output says why: pip put the programs in a folder that is not on the PATH.

```
, localstack, awscli-local
  ____
2/22 [tabulate] WARNING: The script tabulate is installed in '/Users/aviral_pustak/Library/Python/3.14/bin' which is not on PATH.
  ____
Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.
WARNING: The script pysemver is installed in '/Users/aviral_pustak/Library/Python/3.14/bin' which is not on PATH.
  ____
Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.
  ____
7/22 [pygments] WARNING: The script pygmentize is installed in '/Users/aviral_pustak/Library/Python/3.14/bin' which is not on PATH.
  ____
Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.
  ____
14/22 [markdown-it-py] WARNING: The script markdown-it is installed in '/Users/aviral_pustak/Library/Python/3.14/bin' which is not on PATH.
  ____
Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.
  ____
16/22 [cffi] WARNING: The script cffi-gen-src is installed in '/Users/aviral_pustak/Library/Python/3.14/bin' which is not on PATH.
  ____
Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.
  ____
20/22 [localstack] WARNING: The script localstack is installed in '/Users/aviral_pustak/Library/Python/3.14/bin' which is not on PATH.
  ____
Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.
Successfully installed asncrypto-1.5.1 awscli-local-0.22.2 cachetools-7.2.1 cffi-2.1.1 cryptography-50.0.2 dnslib-0.9.26 docker-7.2.0 localstack-2026.8.2 localstack-client-2.12 markdown-it-py-4.2.0 mdurl-0.1.2 packaging-26.3 plux-1.16.0 putils-7.2.2 pycparser-3.0 pygments-2.21.0 pyjwt-2.15.1 pyotp-2.10.0 pyyaml-6.0.3 rich-15.0.0 semver-3.1.0 tabulate-0.10.0

[notice] A new release of pip is available: 26.1.2 -> 26.2.1
[notice] To update, run: pip install --upgrade pip
aviral_pustak@Rohans-MacBook-Pro ~ % /tmp % localstack --version
zsh: command not found: localstack
aviral_pustak@Rohans-MacBook-Pro ~ % /tmp % localstack --version
zsh: command not found: localstack
aviral_pustak@Rohans-MacBook-Pro ~ % /tmp %
```

Screenshot 5: pip warns that ~/Library/Python/3.14/bin is not on PATH; localstack is 'not found'

Fix: add that folder to the PATH in ~/.zshrc and reload it.

```
echo 'export PATH="$HOME/Library/Python/3.14/bin:$PATH"' >> ~/.zshrc
source ~/.zshrc
```

Result

```
aviral_pustak@Rohans-MacBook-Pro ~ % grep "Library/Python" ~/.zshrc
export PATH="$HOME/Library/Python/3.14/bin:$PATH"
aviral_pustak@Rohans-MacBook-Pro ~ % echo 'export PATH="$HOME/Library/Python/3.14/bin:$PATH"' >> ~/.zshrc
aviral_pustak@Rohans-MacBook-Pro ~ % source ~/.zshrc
aviral_pustak@Rohans-MacBook-Pro ~ % for t in brew git docker kubectl kind minikube helm terraform tfint aws python3 node npm jq yq make trivy hadolint localstack ansible; do
if command -v $t >/dev/null 2>&1; then
    printf "%-12s OK    %s\n" "$t" "$t"
else
    printf "%-12s MISSING\n" "$t"
fi
done
brew      OK    /usr/local/bin/brew
git       OK    /usr/bin/git
docker   OK    /usr/local/bin/docker
kubectl  OK    /usr/local/bin/kubectl
kind      OK    /usr/local/bin/kind
minikube  OK    /usr/local/bin/minikube
helm      OK    /usr/local/bin/helm
terraform OK    /usr/local/bin/terraform
tfint     OK    /usr/local/bin/tfint
aws       OK    /usr/local/bin/aws
python3   OK    /Library/Frameworks/Python.framework/Versions/3.14/bin/python3
node      OK    /usr/local/bin/node
npm       OK    /usr/local/bin/npm
jq        OK    /usr/bin/jq
yq        OK    /usr/local/bin/yq
make      OK    /usr/bin/make
trivy     OK    /usr/local/bin/trivy
hadolint  OK    /usr/local/bin/hadolint
localstack OK    /Users/aviral_pustak/Library/Python/3.14/bin/localstack
ansible   OK    /Users/aviral_pustak/Library/Python/3.14/bin/ansible
aviral_pustak@Rohans-MacBook-Pro ~ %
```

Screenshot 6: PATH fix loaded, and the same check now shows all 20 tools OK

Cleaning up

The source builds left several large build tools behind. Two commands removed them and freed 13.6 GB.

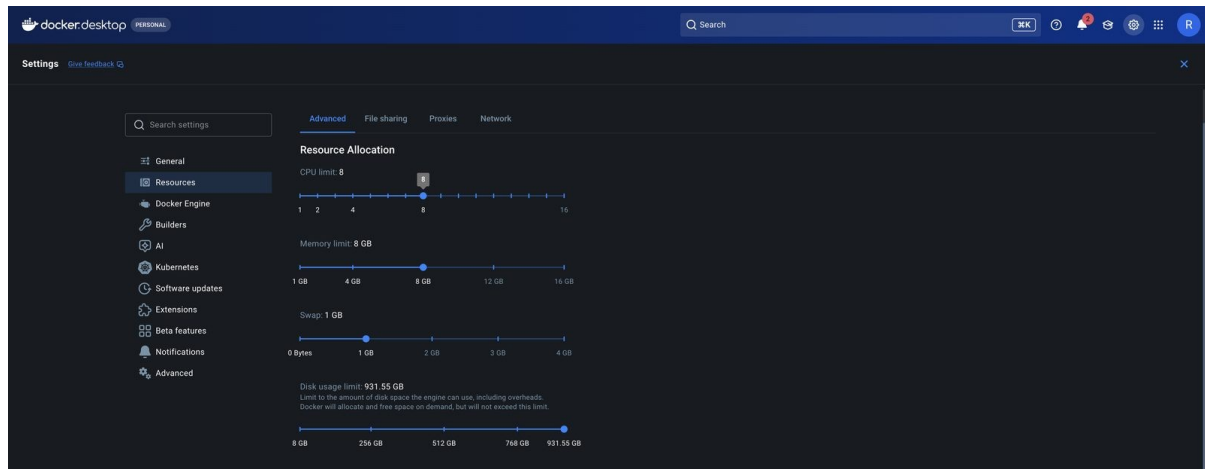
```
brew autoremove
brew cleanup --prune=all
df -h /
```

```
Removing: /Users/aviral_pustak/Library/Logs/Homebrew/automake... (5 files, 76.6KB)
Removing: /Users/aviral_pustak/Library/Logs/Homebrew/peg-turbo... (11 files, 2MB)
Removing: /Users/aviral_pustak/Library/Logs/Homebrew/trivy... (4 files, 288.4KB)
Removing: /usr/local/var/homebrew/tmp/cellar/tree... (8 files, 156.6KB)
Removing: /usr/local/var/homebrew/tmp/cellar/naturin... (30 files, 29.2MB)
Removing: /usr/local/var/homebrew/tmp/cellar/pycparser... (88 files, 266.9KB)
Removing: /usr/local/var/homebrew/tmp/cellar/libyaml... (9 files, 338.3KB)
Removing: /usr/local/var/homebrew/tmp/cellar/cffi... (70 files, 1.2MB)
Removing: /usr/local/var/homebrew/tmp/cellar/openjdk@17... (434 files, 319.3MB)
Removing: /usr/local/var/homebrew/tmp/cellar/certifi... (56 files, 20.6KB)
Removing: /usr/local/var/homebrew/tmp/cellar/libiodm... (87 files, 1.3MB)
Removing: /usr/local/var/homebrew/tmp/cellar/libssh... (24 files, 1.6MB)
Pruned 0 symbolic links and 83 directories from /usr/local
Removing: /usr/local/Homebrew/Library/Homebrew/vendor/portable-ruby/3.4.8... (1,584 files, 36MB)
Removing: /usr/local/Homebrew/Library/Homebrew/vendor/portable-ruby/4.0.5.1... (1,787 files, 34.6MB)
Removing: /usr/local/Homebrew/Library/Homebrew/vendor/portable-ruby/4.0.6... (1,787 files, 34.6MB)
=> This operation has freed approximately 13.6GB of disk space.
aviral_pustak@Rohans-MacBook-Pro ~ % df -h /
Filesystem      Size      Used Avail Capacity iused ifree %used    Mounted on
/dev/disk1s2s1  932Gi    120Gi   812Gi     13%    120 459K    0%    /
```

Screenshot 7: Cleanup freed approximately 13.6 GB; 342 GB available

4. Step 3: Docker Resources

Local Kubernetes clusters run inside Docker, so Docker Desktop needs enough memory. On a 16 GB machine I gave it 8 GB, which leaves half for macOS.



Screenshot 8: Docker Desktop > Settings > Resources: 8 CPUs, 8 GB memory

5. Step 4: A Three-Node Test Cluster with kind

A plain kind create cluster gives a single node. To get one control plane and two workers, kind needs a small config file.

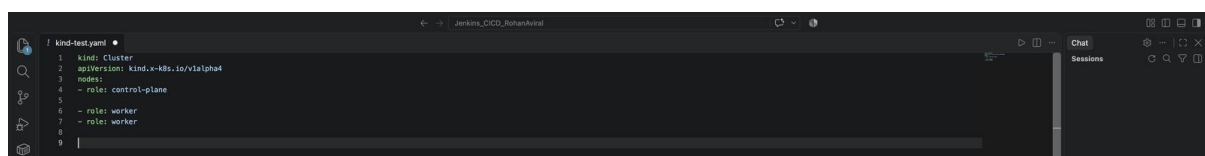
```
# kind-test.yaml
kind: Cluster
apiVersion: kind.x-k8s.io/v1alpha4
nodes:
- role: control-plane
- role: worker
- role: worker
```

Problem 4: an example that pinned a 2019 Kubernetes version

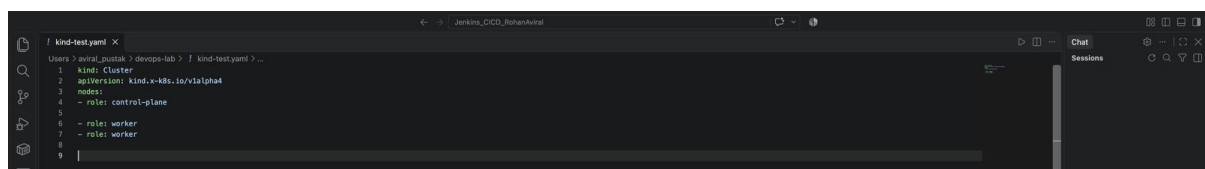
The example I first found in the kind documentation had an image: line on every node, pinning kindest/node v1.16.4, and only one worker. That example exists to show how to pin a version. I deleted the image: lines so kind uses the version it was built and tested with, and added a second worker.

Problem 5: 'no such file or directory'

The first two attempts to create the cluster failed because the file was open in VS Code but had never been saved. The white dot on the tab is the sign.

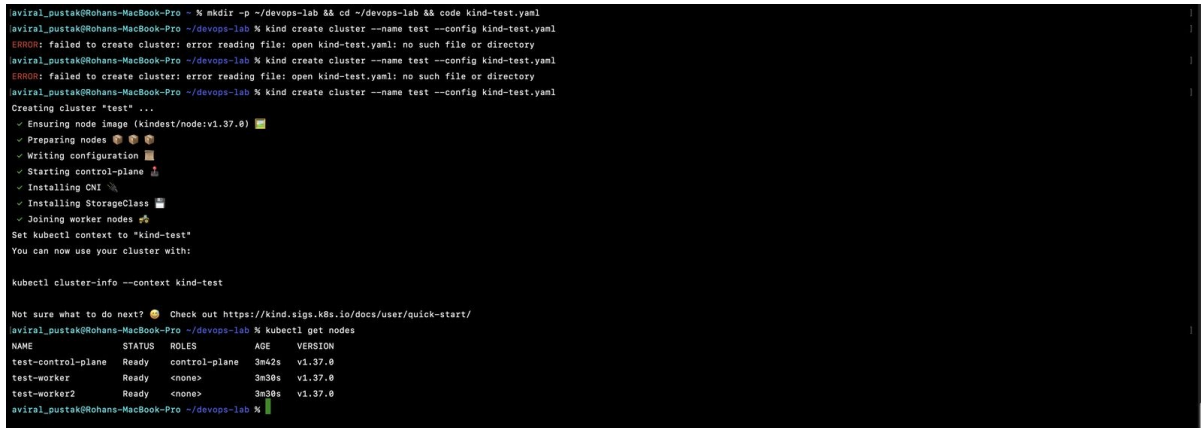


Screenshot 9: The dot next to the file name means unsaved changes



Screenshot 10: After Cmd + S the dot becomes a cross and the file exists on disk

```
kind create cluster --name test --config kind-test.yaml
kubectl get nodes
kind delete cluster --name test
```



```
aviral_pustak@Rohans-MacBook-Pro ~ % mkdir -p ~/devops-lab && cd ~/devops-lab && code kind-test.yaml
aviral_pustak@Rohans-MacBook-Pro ~/devops-lab % kind create cluster --name test --config kind-test.yaml
ERROR: failed to create cluster: error reading file: open kind-test.yaml: no such file or directory
aviral_pustak@Rohans-MacBook-Pro ~/devops-lab % kind create cluster --name test --config kind-test.yaml
ERROR: failed to create cluster: error reading file: open kind-test.yaml: no such file or directory
aviral_pustak@Rohans-MacBook-Pro ~/devops-lab % kind create cluster --name test --config kind-test.yaml
Creating cluster "test" ...
 ✓ Ensuring node image (kindest/node:v1.37.0)
 ✓ Preparing nodes
 ✓ Writing configuration
 ✓ Starting control-plane
 ✓ Installing CNI
 ✓ Installing StorageClass
 ✓ Joining worker nodes
Set kubectl context to "kind-test"
You can now use your cluster with:

kubectl cluster-info --context kind-test

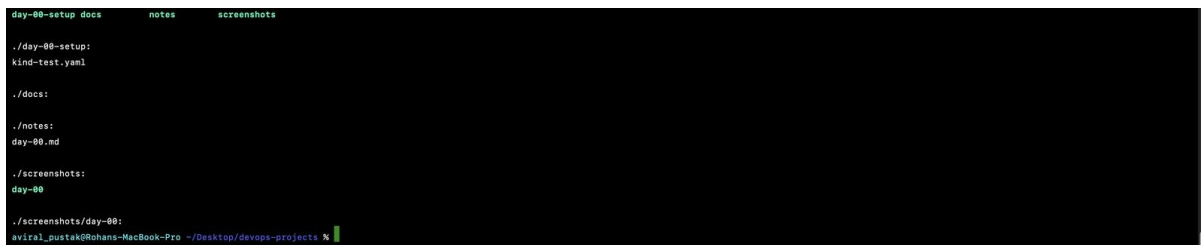
Not sure what to do next? Check out https://kind.sigs.k8s.io/docs/user/quick-start/
aviral_pustak@Rohans-MacBook-Pro ~/devops-lab % kubectl get nodes
NAME                STATUS    ROLES    AGE   VERSION
test-control-plane   Ready    control-plane   3m42s   v1.37.0
test-worker          Ready    <none>         3m38s   v1.37.0
test-worker2         Ready    <none>         3m38s   v1.37.0
aviral_pustak@Rohans-MacBook-Pro ~/devops-lab %
```

Screenshot 11: Two failed attempts, then the cluster is created: three nodes Ready on Kubernetes v1.37.0

I deleted the cluster afterwards. Local clusters use memory while idle, so the habit is create, use, delete.

6. Step 5: Workspace

All 20 projects live under one folder on the Desktop, with shared folders for notes, screenshots and documents.



```
day-00-setup docs  notes  screenshots

./day-00-setup:
kind-test.yaml

./docs:

./notes:
day-00.md

./screenshots:
day-00

./screenshots/day-00:
aviral_pustak@Rohans-MacBook-Pro ~/Desktop/devops-projects %
```

Screenshot 12: ~/Desktop/devops-projects after organising the Day 0 files

7. What I Learned

- On an Intel Mac, prefer release binaries, pip or Docker images over Homebrew for tools that would compile from source.
- 'Command not found' straight after a successful install is usually a PATH problem. Read the installer's warnings.
- 'No such file or directory': run ls and pwd before retrying the same command.
- Before copying an example from documentation, check which lines are needed and look at any version numbers in it.
- Check what is installed before installing anything.

Commands worth remembering

Command	What it does
command -v <tool>	Is this tool installed, and where

<code>brew search <word></code>	Find the right package name
<code>df -h /</code>	Free disk space
<code>echo \$PATH</code>	Folders the shell searches for commands
<code>source ~/.zshrc</code>	Reload shell configuration
<code>chmod +x <file></code>	Make a downloaded binary executable
<code>ls -la</code>	List files, including hidden ones
<code>kubectl get nodes</code>	List the nodes of the current cluster

Next: Project 1, Linux Ops Toolkit.